DEEP LEARNING · FINAL PROJECT

English → Khmer Neural Machine Translation

Training from scratch vs. a frozen pretrained backbone vs. full fine-tuning

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I want to eat papaya salad.

ខ្ញុំចង់ញ៉ាំបុកល្ហុង។

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1 · PROBLEM DEFINITION

Translate one English sentence into Khmer

INPUT · ENGLISH (LATIN SCRIPT)

I want to eat papaya salad.



OUTPUT · KHMER (KHMER SCRIPT)

ខ្ញុំចង់ញ៉ាំបុកល្ហុង។

Task type: **sequence-to-sequence generation**, a variable-length text in and a variable-length text out, one sentence at a time.

Low-resource language

About 17 million speakers, but little parallel English–Khmer text exists for training.

No spaces between words

Khmer script has no reliable word boundaries, so word-level tools and metrics break.

Research question

On a free Colab GPU, which training strategy translates best, and at what cost?

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2 · DATASET

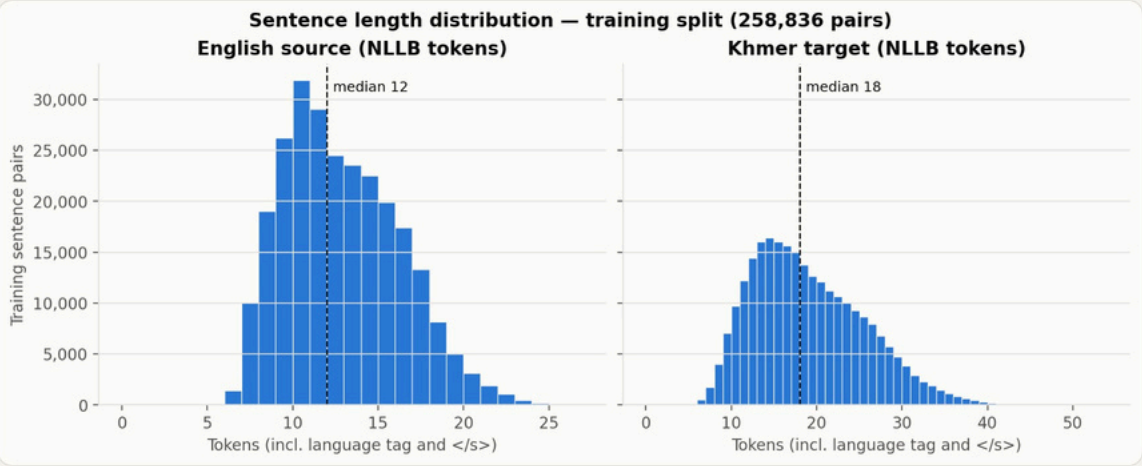
SeyhaLite: 323,545 English–Khmer pairs

SeyhaLite/Translate-English-Khmer-All · Hugging Face · Apache-2.0 · 366,174 raw pairs, 11.6% removed by cleaning · split 80 / 10 / 10 with seed 42

258,836
training pairs

32,354
validation pairs

5,000
test sentences, identical for every approach (of 32,355)



Known biases and noise

- Templated: only 4,102 English words
- 3.1% of test references also in training
- Colloquial spelling: ញុំ for ខ្ញុំ (“I”)
- Spacing: 73% of references use spaces

2 · PIPELINE

One pipeline, identical for every approach

1 · Load
Hugging Face dataset, 366k pairs

→

2 · Clean
strip, drop empty, de-duplicate English

→

3 · Split
80 / 10 / 10, fixed seed 42

→

4 · Tokenize
A: own SentencePiece; B, C: NLLB tokenizer

→

5 · Train
A, B, C; validation for all selection

→

6 · Evaluate
same 5,000 sentences, beam 4, chrF++

Same test-time input

Every model gets the same raw English sentence and is scored against the same raw Khmer reference. Only the tokenizer differs: each model reads its own vocabulary.

No leakage

Duplicate English sentences are removed before splitting. The test set is used only for final numbers; tuning and early stopping use validation.

Reproducible

Seeds fixed in Python, NumPy and PyTorch. torch.save checkpoints with --resume survive Colab disconnects.

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3 · APPROACHES

Three training strategies on the same task

A · FROM SCRATCH

Transformer encoder-decoder

11.5M

parameters, all trained from random init

4 + 4 layers, d_model 256

Own 16k SentencePiece vocabulary

Why: the no-transfer baseline: what the data alone can teach.

B · FROZEN BACKBONE

NLLB-200-distilled-600M

50.4M

of 615M parameters trained (8.2%)

Only decoder cross-attention trained

All other weights stay pretrained

Why: parameter-efficient transfer learning.

C · FULL FINE-TUNING

NLLB-200-distilled-600M

615.1M

parameters, all trained

Adafactor + gradient checkpointing

Fits an 8 GB GPU (peak 6.7 GB)

Why: maximum capacity to adapt to the data.

Reference only: zero-shot NLLB-600M, pretrained but never trained on this data. The three differ in training strategy (scratch → frozen → full) and in architecture size (11.5M vs. 615M).

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Which parts of the Transformer are trained

A · scratch (own 11.5M model)	B · frozen backbone	C · full fine-tuning	
Trained · random init	Frozen · pretrained	Trained · from pretrained	Embeddings + output layer (tied)
Trained · random init	Frozen · pretrained	Trained · from pretrained	Encoder: reads the English
Trained · random init	Frozen · pretrained	Trained · from pretrained	Decoder self-attention + feed-forward
Trained · random init	Trained · 50.4M	Trained · from pretrained	Decoder cross-attention: looks at the English

Cross-attention is where the decoder reads the encoded English sentence. Re-training only its 12 blocks re-aligns English → Khmer, while 92% of NLLB's weights keep their knowledge of 200 languages.

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Same test, same metrics, fixed budgets

chrF++ (primary)

Character n-grams plus word bigrams. Suits Khmer, which has no reliable word boundaries.

chrF

Character-only and whitespace-insensitive: unaffected by inconsistent Khmer spacing.

spBLEU

BLEU on SentencePiece pieces (sacreBLEU flores200 tokenizer), as in the NLLB paper.

Approach	Optimizer and key settings	Budget	GPU	Train time
A	AdamW · lr 1e-3 · label smoothing 0.1	15 epochs (30,345 steps)	Tesla T4	21.8 min
B	AdamW · lr 1e-4	16,000 steps (≈ 1 epoch)	RTX 4070 Laptop	61.6 min
C	Adafactor · lr 1e-4 · grad. checkpointing	2,000 steps	Tesla T4	27.5 min

Test: the same 5,000 held-out sentences, beam search with 4 beams · dropout 0.1 · fp16 · effective batch 128 (A), 16 (B, C) · train time = optimizer steps only

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Grid search on validation: learning rate × dropout

Dropout	A · 3e-4	A · 5e-4	A · 1e-3	C · 1e-5	C · 3e-5	C · 1e-4
0.1	89.17	93.59	96.73 best	47.78	62.89	80.14 best
0.3	65.36	83.67	92.07	43.28	51.88	66.26

Validation chrF++ per short trial · A: 3 epochs on an 80k-pair subset · C: 300 steps · 6 trials each · weight decay 0.01

Higher learning rate wins

Best at the top of both grids: A 1e-3, C 1e-4 (5× the usual 2e-5). The true optimum may be higher.

Dropout 0.3 is always worse

−4.5 to −23.8 chrF++. The models underfit at these budgets, so extra regularization only slows learning.

Budget matters most

B trained 4× longer (4,000 → 16,000 steps): test chrF++ 93.82 → 99.31. The first run was undertrained.

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All three approaches far exceed zero-shot

Approach	chrF++	chrF	BLEU	Trainable	Train time	Peak VRAM
Zero-shot NLLB (reference)	40.53	50.31	13.69	0	–	–
A: Transformer from scratch	99.13	99.16	98.86	11.5M	21.8 min	1.6 GB
B: NLLB frozen backbone (best)	99.31	99.45	99.21	50.4M	61.6 min *	5.2 GB
C: NLLB full fine-tuning	98.42	98.71	98.09	615.1M	27.5 min	6.7 GB

Same 5,000 in-domain test sentences · beam search, 4 beams · BLEU = spBLEU · * B on an RTX 4070 Laptop GPU, the others on a Tesla T4

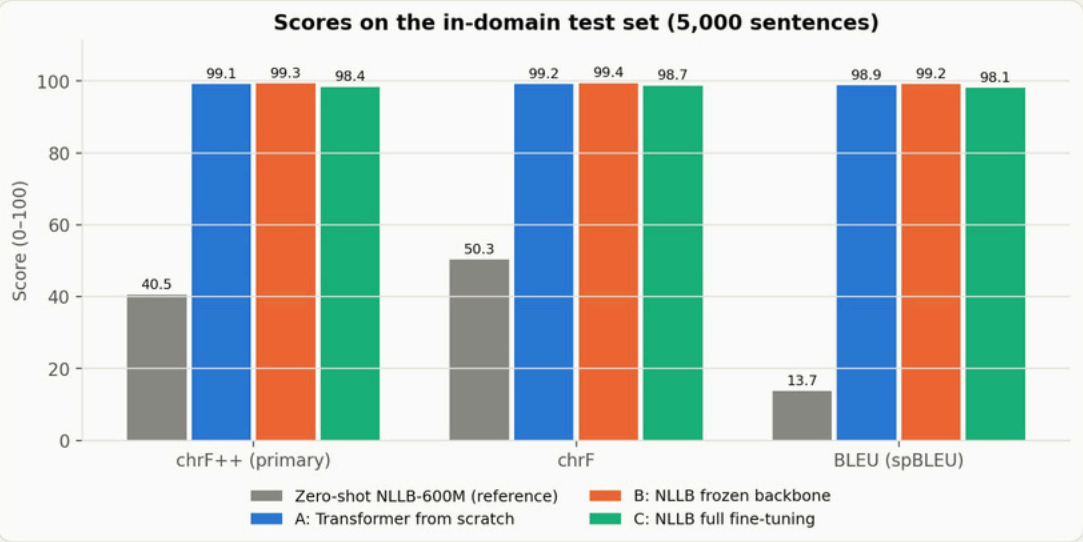
+58 to +59
chrF++ over zero-shot for every trained approach

0.9
chrF++ separates the top three

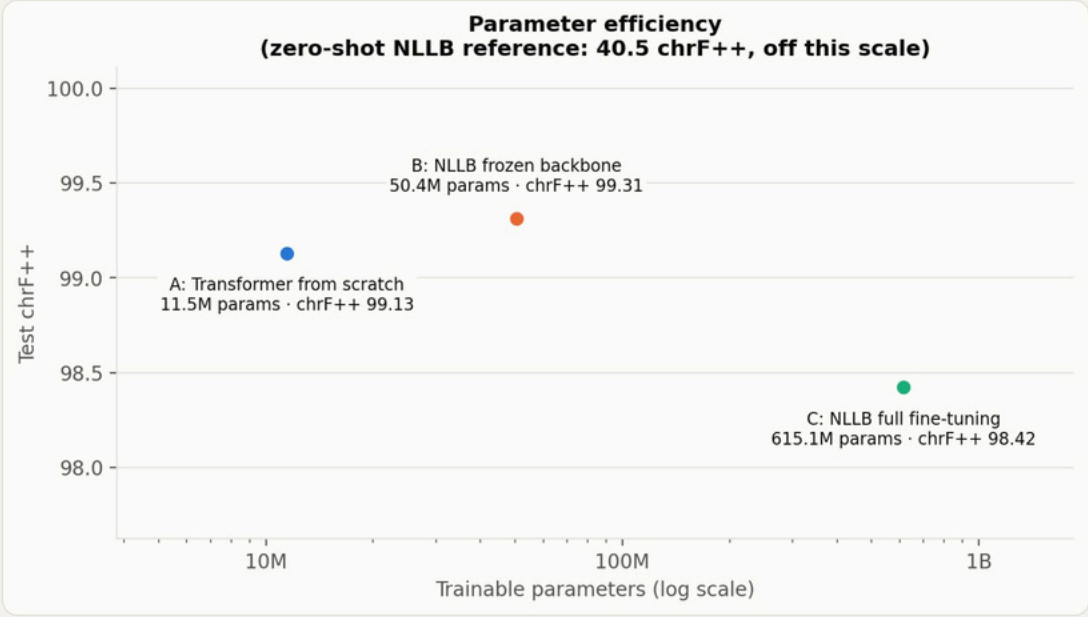
32k
training pairs were all C needed to reach 98.4

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Test scores and parameter efficiency



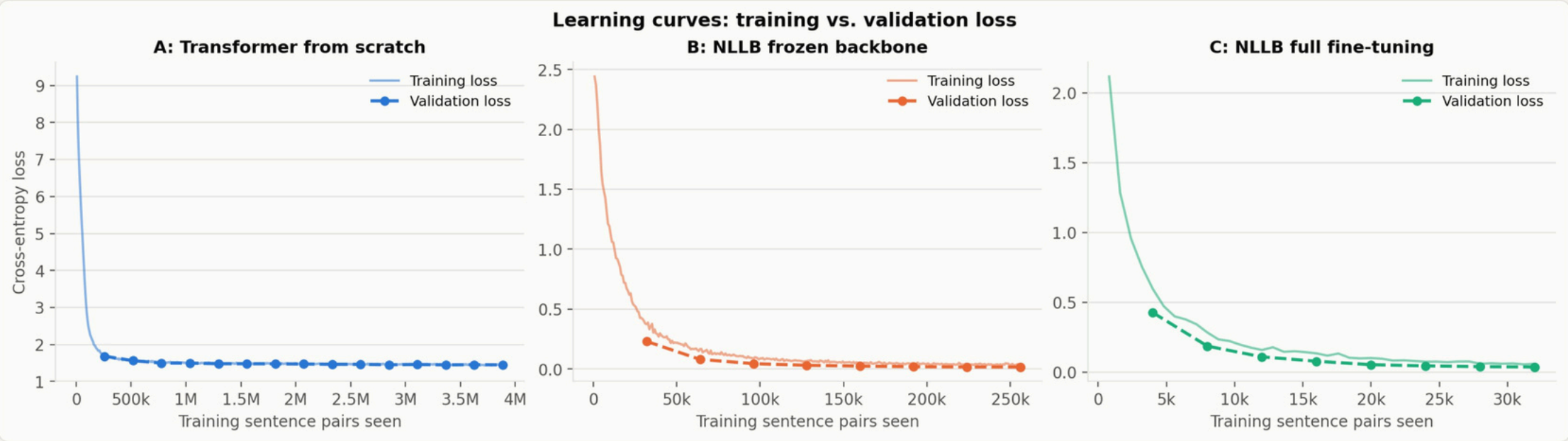
Test chrF++, chrF and BLEU per approach (5,000 sentences)



Test chrF++ vs. trainable parameters (log scale): B scores highest while training 8.2% of NLLB

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Training vs. validation loss: no overfitting



A · scratch

Validation loss 1.68 → 1.45, still improving at epoch 15; early stopping never triggered.

B · frozen backbone

4,000-step run still rising: underfitting. Retrained for 16,000 steps; plateau at 99.5.

C · full fine-tuning

Converges within 2,000 steps; most sample-efficient: 90 chrF++ after only 8k pairs.

What still goes wrong, and why

	Zero-shot	A	B	C
Exact match	0.0%	95.8 %	97.0 %	92.5 %
Partly correct	46.4%	3.3%	2.6%	6.5%
Failure (chrF++ < 40)	51.9%	0.9%	0.3%	0.8%

Share of the 5,000 test sentences. Categories are assigned by rules in src/error_analysis.py.

1 · Register and spacing mismatch (zero-shot)

“My boyfriend is angry.” → មិត្ត ប្រុស របស់ ខ្ញុំ មាន កំហឹង; reference សង្សារកំពុងខឹង. After training, B outputs the reference exactly.

2 · Attention drift (zero-shot)

On a long formal sentence the decoder loops, repeating និស្សិត (“student”) until the length limit.

3 · Rare-word breakdown (A only)

“Surprising my mom with a gift.” → ការចេញ. A has no pretrained knowledge to fall back on.

4 · Reference noise

Only 6 of 5,000 fail for all three. “It is raining.”: B and C say ភ្លៀងកំពុងធ្លាក់ (correct); the reference is ភ្លៀងហើយ.

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MY OWN QUALITATIVE CHECK

99 chrF++ on the test set, yet everyday sentences failed

MODEL A · EXPERIMENT 1

i want to study english in my parents school

ឪពុកម្តាយក្រុមទាមទាញ្ចម។

Unreadable Khmer

MODEL B · EXPERIMENT 1

i want to kiss you

ញ្ចមចង់ផ្កាឯង។

“I want flower you”

The test set is templated (4,102 English words), so the high scores measured fit to the templates, not general translation.

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Add varied data, change nothing else

258,836
SeyhaLite training pairs: conversational, templated

+

17,845 × 3
ALT news pairs (Asian Language Treebank), professionally translated, formal Khmer, CC-BY-4.0

=

312,371
training pairs, shuffled; ALT is about 17%

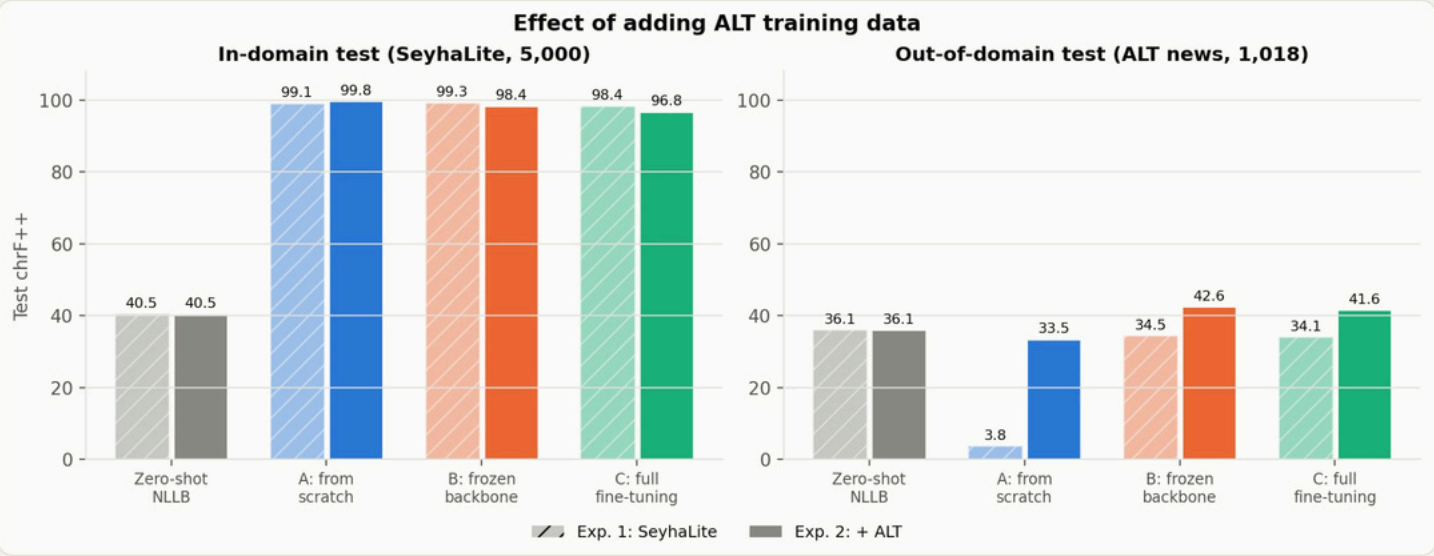
Two test sets

1. The same 5,000 SeyhaLite sentences (in-domain).
2. ALT's 1,018 news sentences (out-of-domain). Experiment 1's models were scored on ALT too.

Fair comparison

Same architectures, tuned hyperparameters, budgets, seed and decoding. Validation: 1,000 SeyhaLite + 1,000 ALT. All three retrained on a Colab T4.

Every model improved on unseen news text



chrF++ per approach · hatched = experiment 1, solid = experiment 2

+29.7

A: 3.8 → 33.5 chrF++ on ALT news

+8.1

B: 34.5 → 42.6, still the best

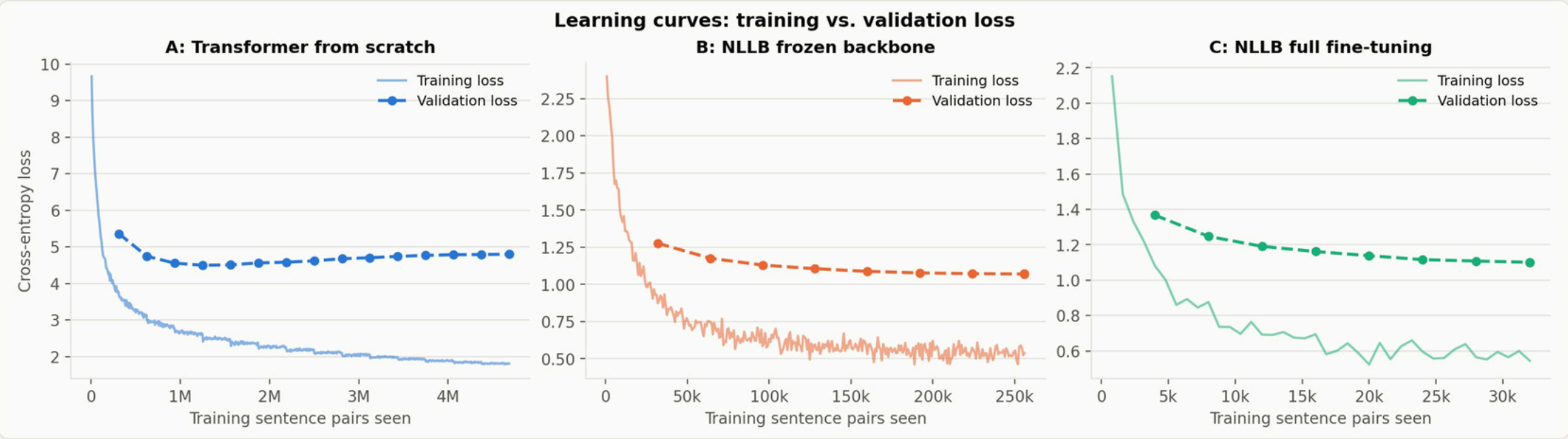
+7.5

C: 34.1 → 41.6

B and C now beat zero-shot NLLB (36.1) on news.
In-domain cost: B −0.9, C −1.6.

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A now overfits; B and C do not



A · overfitting

Validation loss is lowest after epoch 4 (4.50), then rises to 4.80 while training loss keeps falling.

Why

Each ALT pair is seen 45 times (3× upsampling × 15 epochs), and the 11.5M model memorises them.

B and C

Validation loss falls throughout (B 1.28 → 1.07, C 1.37 → 1.10): no overfitting within budget.

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Why B wins, in course concepts

Transfer learning

B keeps NLLB's knowledge of 200 languages: 92% of its weights stay frozen. Training only cross-attention re-aligns English → Khmer: exact matches 0.04% → 97.0%.

Freezing as regularization

Few trainable parameters adapt to the corpus without overwriting prior knowledge. B is best in-domain (99.31) and out of domain (42.6).

Capacity, inductive bias and data size

A (11.5M, no prior) learns the templates from 3.88M pair-passes, but breaks on rare words and overfits the small ALT set.

Sample efficiency vs. cost

C learns most per example (98.4 after 32k pairs) but updates 615M parameters: 0.83 s/step vs. 0.20 s/step for B on a T4.

Best quality → **B** | Least data and time → **C** | Smallest and fastest → **A** (46 MB, 8 s for 5,000 sentences)

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What this study cannot claim

Templated in-domain test

99+ scores measure fit to the templates. ALT adds one out-of-domain set, but only news.

Vocabulary gaps

“hug” and “kiss” occur 0 times in 259k SeyhaLite pairs; every trained model still mistranslates them.

Unequal budgets

A saw 3.88M pair-passes, B 256k, C 32k: the ranking holds at these budgets, not at equal training.

No statistics

One seed per run and no confidence intervals; the top three differ by less than 1 chrF++.

Single reference

Automatic metrics only; valid alternative translations can be scored as wrong.

Experiment 2 not re-tuned

ALT upsampling (3×) was not tuned, and A overfits the repeated ALT pairs.

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Takeaways and next steps

Key takeaways

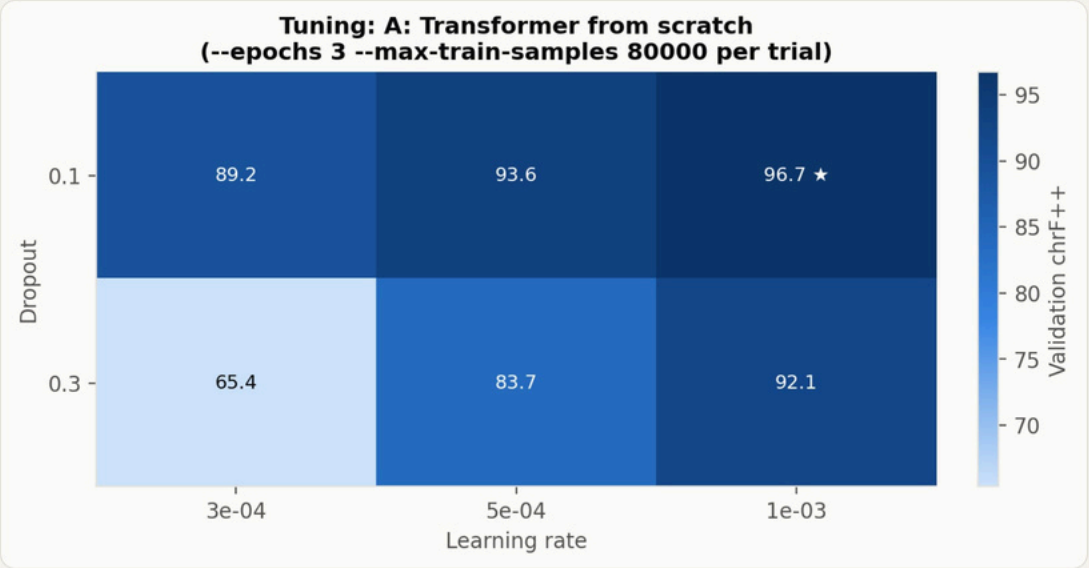
1. Pretrained model + light adaptation wins: B is best in-domain (99.31) and on news (42.6), training only 8.2% of the weights.
2. A high test score is not generalisation: the templated test hid A's failures; an out-of-domain test exposed them.
3. Data coverage matters as much as the model: ALT lifted every approach (+7.5 to +29.7), but words absent from all data stay wrong.

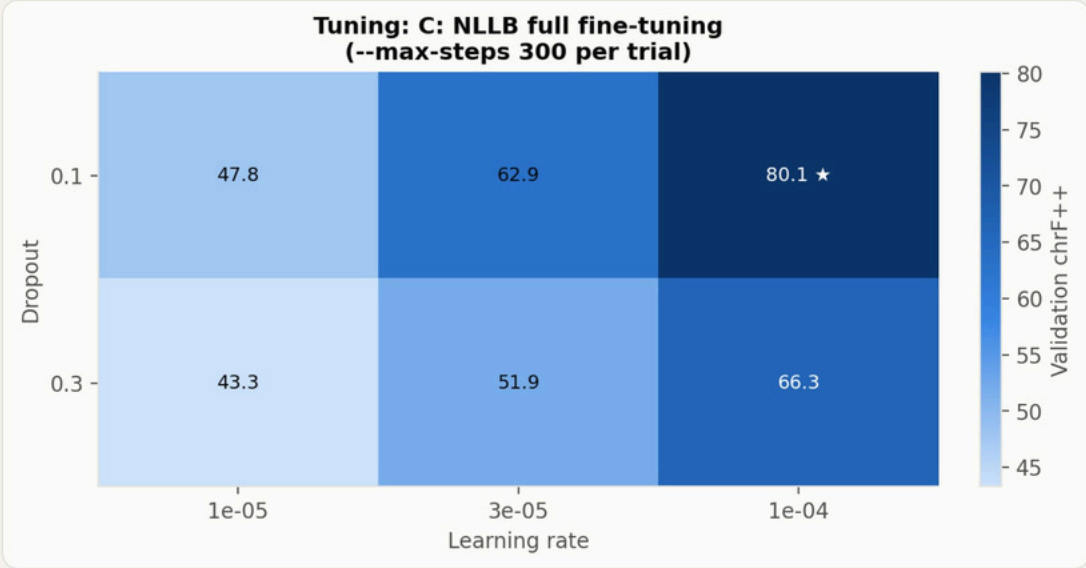
Code, results and live demo: github.com/VimeanLav/english-khmer-translation

Future work, by priority

1. Conversational parallel data (e.g. OPUS Tatoeba)
2. FLORES-200 benchmark: more domains
3. Tune experiment 2: upsampling, early stopping
4. LoRA as a fourth approach
5. Several seeds and confidence intervals

Tuning grids and the training-budget ablation





B budget ablation: 4,000 steps → test chrF++ 93.82 (75.1% exact matches) · 16,000 steps → 99.31 (97.0% exact matches). B was not tuned for learning rate or dropout.

APPENDIX B

Qualitative probe: everyday sentences

English	A · experiment 1	A · experiment 2	B · experiment 2
i want to hug you	ករាមេញ្ញាមេញ។	ខ្ញុំចង់ហ្សៀង។	ញ្ញមចង់យកចិត្តទុកដាក់ឯង
i want to kiss you	ចង់ទៅទំពិត្រប់ស្លូ។	ខ្ញុំចង់អរគុណ។	ញ្ញមចង់បានអ្នក
Where is the nearest hospital?	តើមន្ទ្រពេទ្យដែលនៅជិតនៅឯណា?	តើមន្ទីរពេទ្យដែលនៅជិតបំផុតនៅឯណា?	មន្ទីរពេទ្យជិតបំផុតនៅណា?
I love my mother.	ញ្ញមស្រលាញ់មេ។	ញ្ញមស្រលាញ់ម្តាយ។	ញ្ញមស្រលាញ់ម៉ាក់។
The weather is very hot today.	អាកាសធាតុនៅក្នុងខែ ក្តៅ គឺ ក្តៅណាស់។	ថ្ងៃនេះ អាកាសធាតុនៅក្នុងខែ ក្តៅណាស់។	ថ្ងៃនេះអាកាសធាតុក្តៅណាស់។

“hug” and “kiss” occur 0 times in SeyhaLite's 259k training pairs. Full probe of 14 sentences: results/augmented/qualitative_probe.md

APPENDIX C

Code map for Q&A

File	What it does
src/data.py	Load, clean, split (seed 42), tokenize; ALT and the experiment-2 mix
src/model_scratch.py	A: Transformer, tied embeddings, sinusoidal positions, beam search
src/train_scratch.py	A: SentencePiece, training loop, torch.save checkpoints, early stopping
src/train_frozen.py	B: freeze everything except decoder cross-attention
src/train_finetune.py	C: full fine-tuning with Adafactor and gradient checkpointing
src/tune.py	Grid search over learning rate × dropout on validation
src/evaluate.py	Identical test evaluation for every approach (beam 4, chrF++)
src/error_analysis.py	Rule-based error categories and failure examples
src/utils/seed.py	Seeds Python, NumPy and PyTorch (torch.manual_seed, CUDA)
notebooks/	Colab runners for experiments 1 and 2 (Drive-backed, resumable)